

Chapter 13 Kinetic theory of gases (气体动理论)



Key words and Terms

Kinetic theory of gases	气体动理论
thermal motion	热运动
Boltzmann constant	玻耳兹曼常量
Maxwell distribution of speeds	麦克斯韦速率分布
most probable speed	最概然速率
root-mean-square speed	方均根速率
average speed	平均速率
mean collision frequency	平均碰撞频率
mean free path	平均自由程
macroscopic quantities	宏观量
microscopic quantities	微观量

Chapter 13 Kinetic theory of gases (气体动理论)

13.1 The ideal gas law

13.2 The molecular interpretation of **pressure** and **temperature**

13.3 Distribution of Molecular speeds

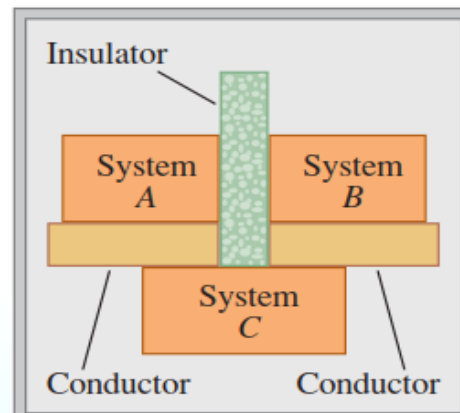
*13.4 Mean free path (平均自由程)

13.1 The ideal gas law

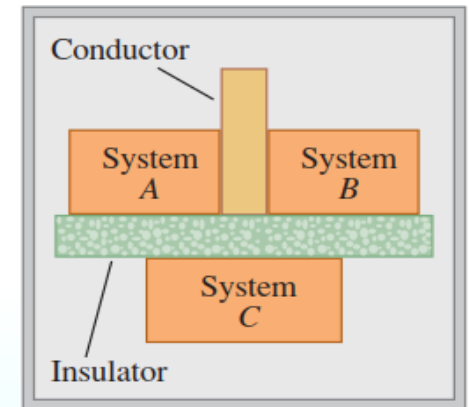
1. The *zeroth* law of thermodynamics

Thermal equilibrium: Thermal equilibrium is a situation in which two objects in **thermal contact** with each other **cease** to exchange **energy** by the process of heat.

If bodies A and B are each in thermal equilibrium with a **third** body C, then A and B are in thermal equilibrium with each other.



If systems A and B are each in thermal equilibrium with system C....

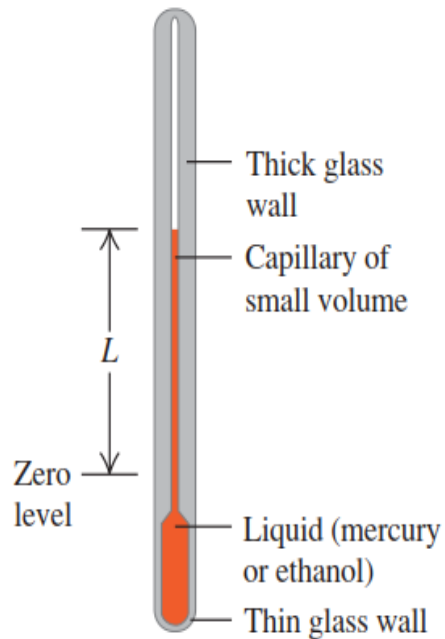


...Then systems A and B are in thermal equilibrium with each other.

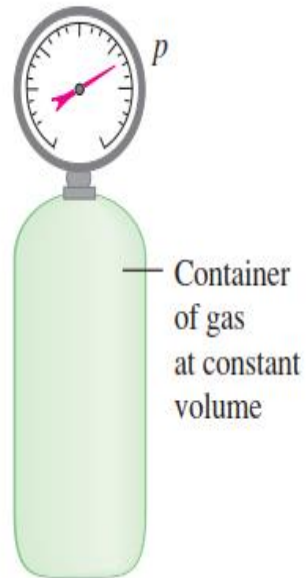
“Every body has a property called temperature.” The zeroth law allows a useful definition of **temperature**.

Three devices for measuring temperature.

(a) Changes in temperature cause the liquid's volume to change.

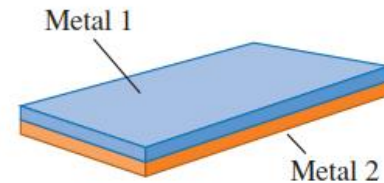


(b) Changes in temperature cause the pressure of the gas to change.

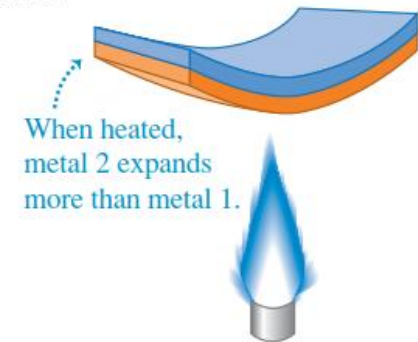


Use of a bimetallic strip as a thermometer.

(a) A bimetallic strip



(b) The strip bends when its temperature is raised.



(c) A bimetallic strip used in a thermometer

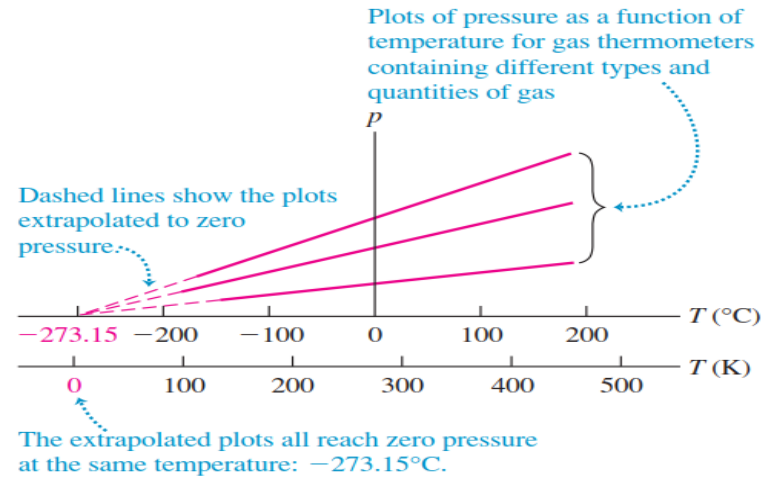


2. Temperature scales(温标)

(a) Kelvin (Absolute temperature) scale, **unit: K**

-273.15⁰C as its zero point

A constant-volume gas thermometer



(b) Celsius scale, **unit: ⁰C**, triple point of water as 0⁰C , steam point as 100⁰C

$$T_C = T_K - 273.15$$

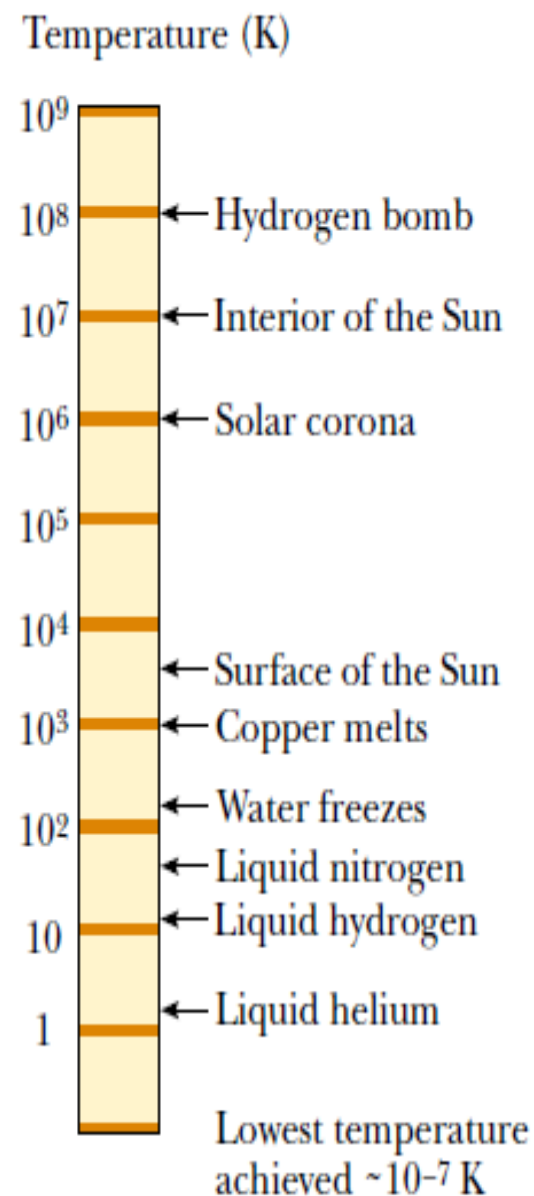
(c) Fahrenheit scale, **unit: ⁰F**, the ice point at 32⁰F and steam point as 212⁰F

$$T_F = \frac{9}{5}T_C + 32^{\circ}F$$

$$\Delta T_C = \Delta T = \frac{5}{9}\Delta T_F$$

Relationships among Kelvin (K), Celsius (C), and Fahrenheit (F) temperature scales.

	K	C	F
Water boils	373	100°	212°
	↑ 100 K	↑ 100 C°	↑ 180 F°
Water freezes	273	0°	32°
CO ₂ solidifies	195	-78°	-109°
Oxygen liquefies	90	-183°	-298°
Absolute zero	0	-273°	-460°



3. Equation of state for an ideal gas (理想气体状态方程)

Equilibrium state(平衡态): A state in which the **macroscopic quantities**(p, V, T) have their unchanged values.

一定量的气体，在不受外界的影响下，经过一定的时间，系统达到一个稳定的 宏观性质不随时间变化的状态.

State variables (P, V, T)

(1)Pressure(P) *unit* : $1 Pa = 1 N \cdot m^{-2}$ $1 atm = 1.013 \times 10^5 Pa$

(2)Volume (V) *unit* : m^3 $1 m^3 = 10^3 L$

(3)Temperature(T) *unit* : K (开尔文)

State variables (p , V , T)

(1) Pressure(p) *unit* : $1 \text{ Pa} = 1 \text{ N} \cdot \text{m}^{-2}$ $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$

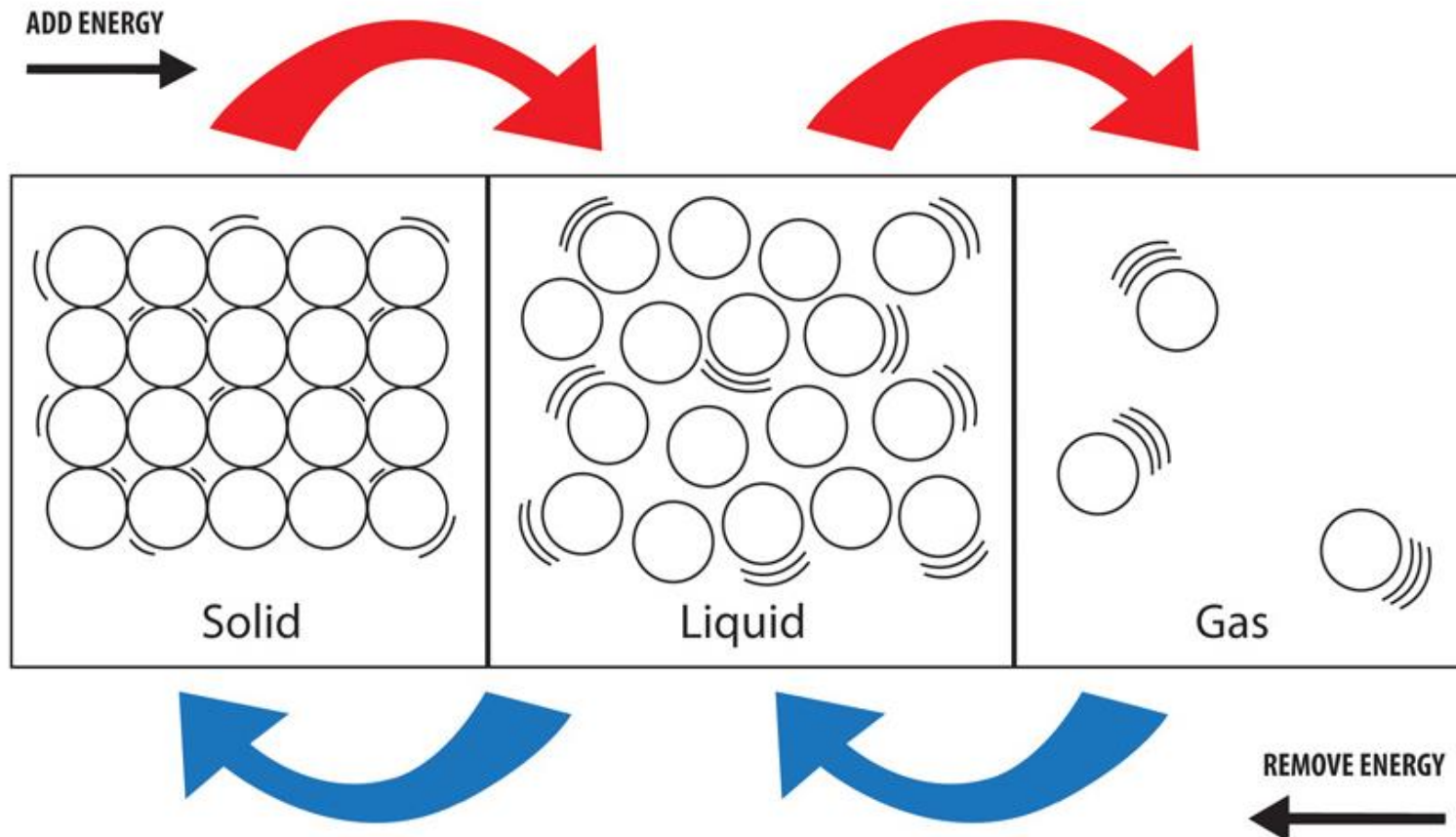


State variables (p , V , T)

(2) Volume (V)

unit : m^3

$1\text{m}^3 = 10^3\text{L}$

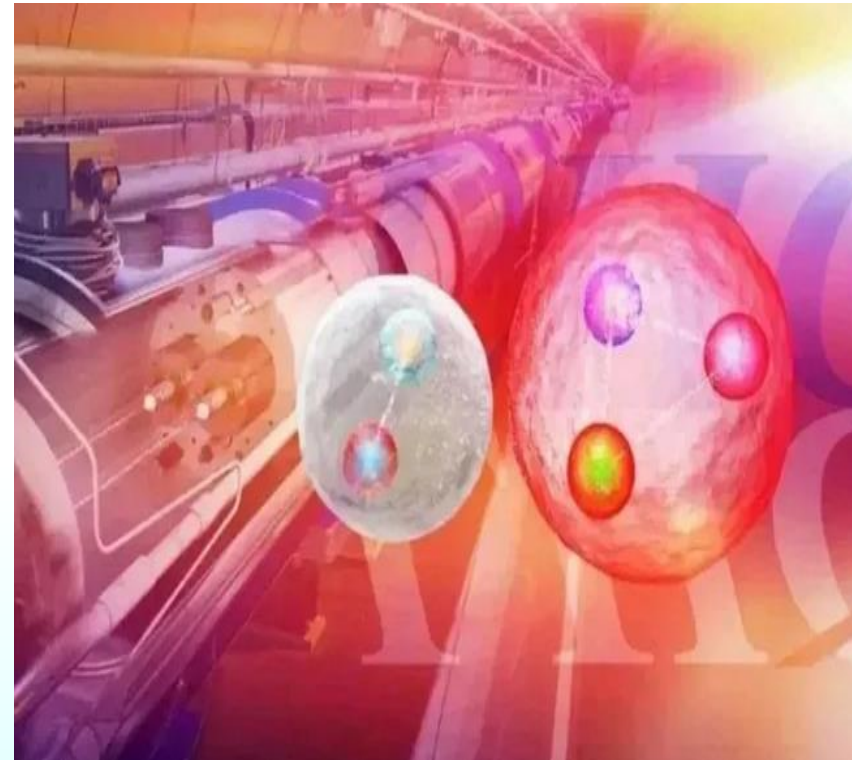
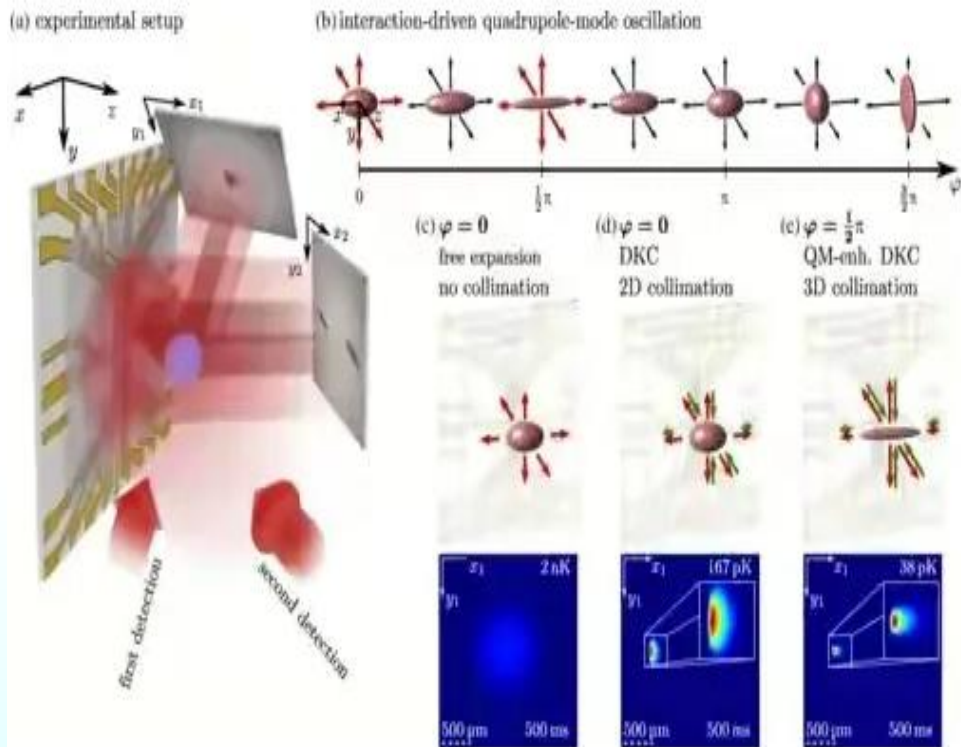


State variables (p, V, T)

(3) Temperature (T) *unit*: K(开尔文)

38皮开尔文

10万亿℃



Ideal gas : at **low densities**, under conditions in which their molecules are **far enough** apart that they do not interact with one another.

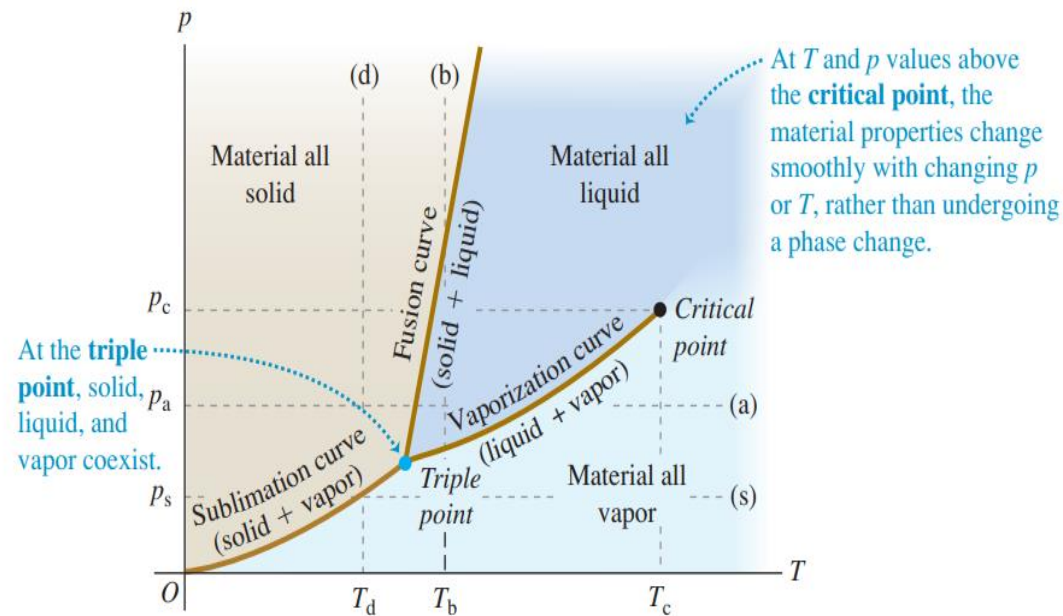
Ideal gas : the pressure is not **too high** and the temperature is **far** from the liquefaction point(液化点)

(**low** pressures and **high** temperatures)

$$pV = \text{constant} (\text{law of Boyle})$$

$$\frac{p}{T} = \text{constant} (\text{law of Charles})$$

$$\frac{V}{T} = \text{constant} (\text{law of Gay-Lussac})$$



Ideal gas : obey the gas laws of Boyle, Charles, and Gray-Lussac

Equation of state for an ideal gas

$$pV = \frac{M}{M_{mol}} RT = nRT \quad (\text{ideal gas law})$$

n : number of moles,

M : mass of ideal gas

$$R = 8.31 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1},$$

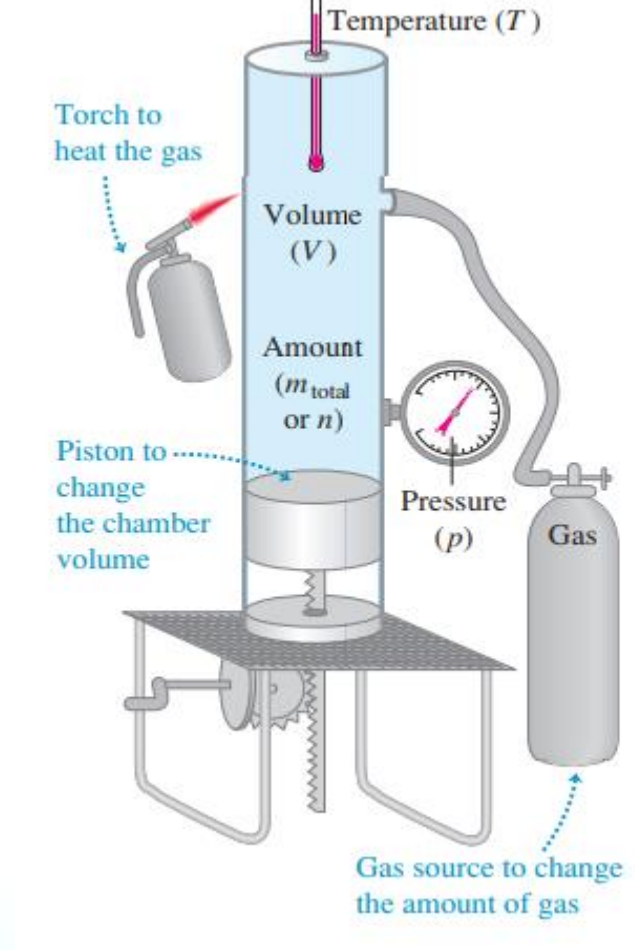
Universal gas constant 普适气体常量

$$pV = nRT = \frac{M}{M_{mol}} RT = \frac{Nm}{N_A m} RT$$

m : mass of a **single** molecule, **N_A** : Avogadro's number

$$pV = N \frac{R}{N_A} T = NkT \quad (\text{differential form})$$

$$k = R / N_A = 1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1} \quad \text{Boltzmann's constant 玻耳兹曼常量}$$



$$nR = Nk$$

* Van der Waals Equation of state (real gas)

$$pV = nRT \quad (\text{ideal gas law})$$

Let ***b*** represent the “**unavailable** volume per mole” of gas.

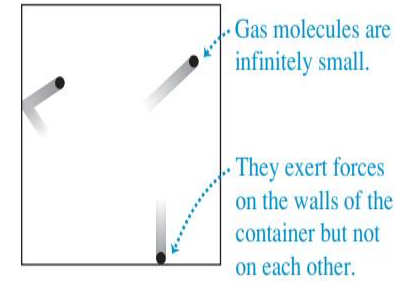
$$p(V - nb) = nRT \Rightarrow P\left(\frac{V}{n} - b\right) = RT$$

We expect the pressure to be reduced by a factor proportional to the density squared ***a*** $(n/V)^2$

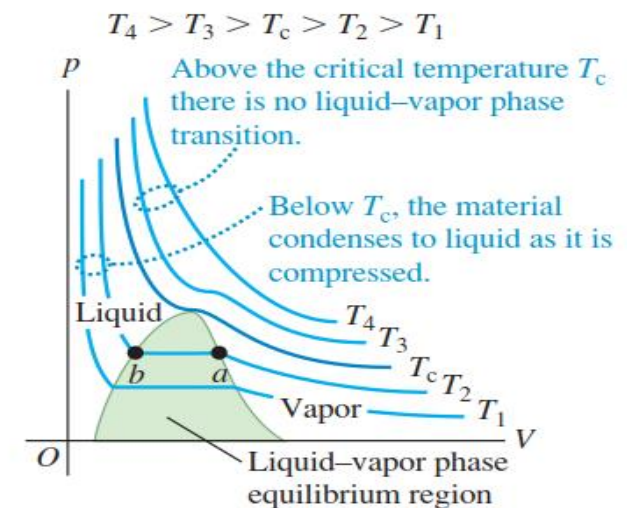
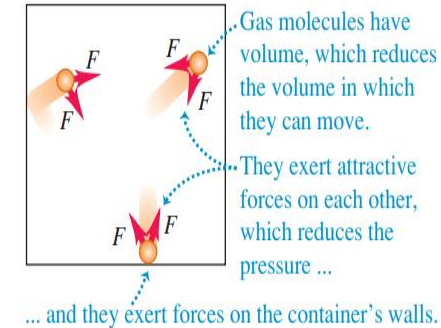
$$p = \frac{RT}{\left(\frac{V}{n} - b\right)} - \frac{a}{(V/n)^2}$$

$$\left(p + \frac{a}{(V/n)^2}\right)\left(\frac{V}{n} - b\right) = RT$$

(a) An idealized model of a gas



(b) A more realistic model of a gas



Question

A sample of an ideal gas is compressed by a piston from 10m^3 to 5m^3 and simultaneously cooled from 273°C to 0°C . As a result there is:

- A. an increase in pressure**
- B. a decrease in pressure**
- C. a decrease in density**
- D. no change in density**
- E. an increase in density**

Question

It is known that 28 g of a certain ideal gas occupy 22.4 liters at **standard conditions** (0°C , 1 atm). The volume occupied by 42 g of this gas at standard conditions is:

- A. 14.9 liters
- B. 22.4 liters
- ☒ C. 33.6 liters
- D. 42 liters
- E. more data are needed

Question

According to the kinetic theory of gases, the pressure of a gas is due to:

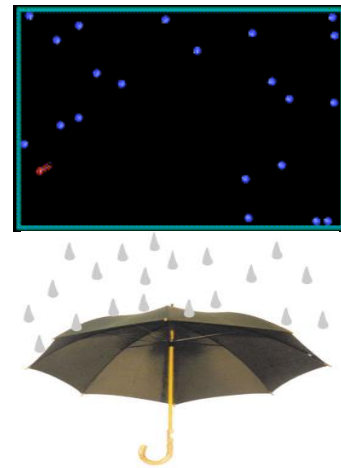
A. change of kinetic energy of molecules as they strike the wall

B. change of momentum of molecules as they strike the wall

C. average kinetic energy of the molecules

D. force of repulsion between the molecules

13.2 The molecular interpretation of pressure and temperature



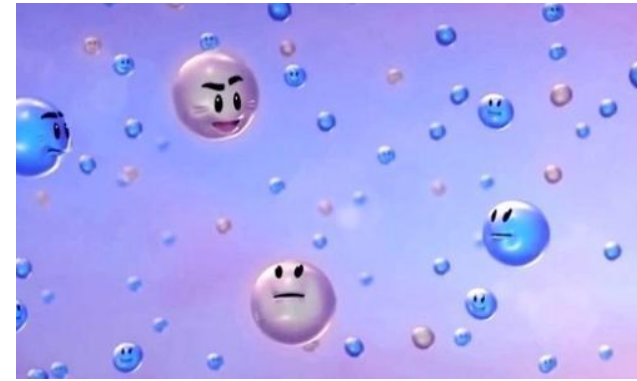
1. The basic postulates of the kinetic theory

- (1) There are a **large number** of molecules moving in **random** directions with a **variety of speeds**.
- (2) The molecules can be treated as **particles**.
- (3) The molecules are assumed to obey the laws of **classical mechanics**, and are assumed to **interact** with one another only when they **collide**.
- (4) Collisions with another molecule or the wall are assumed to be **elastic**.

The actual force due to one molecule is intermittent, but because a huge number of molecules are striking the wall per second, the force is, on average, nearly constant.

单个分子碰撞特性：偶然性、不连续性.

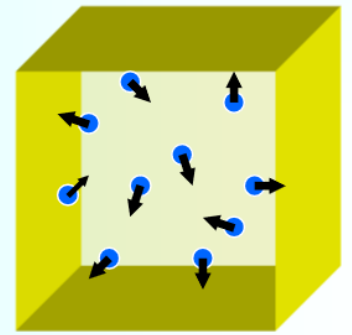
大量分子碰撞的总效果：恒定的、持续的力的作用.



We shall now show that such large-scale properties can be described on a microscopic scale, where matter is treated as a collection of molecules.

The velocity of a molecule

$$\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$$



square of the equation

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

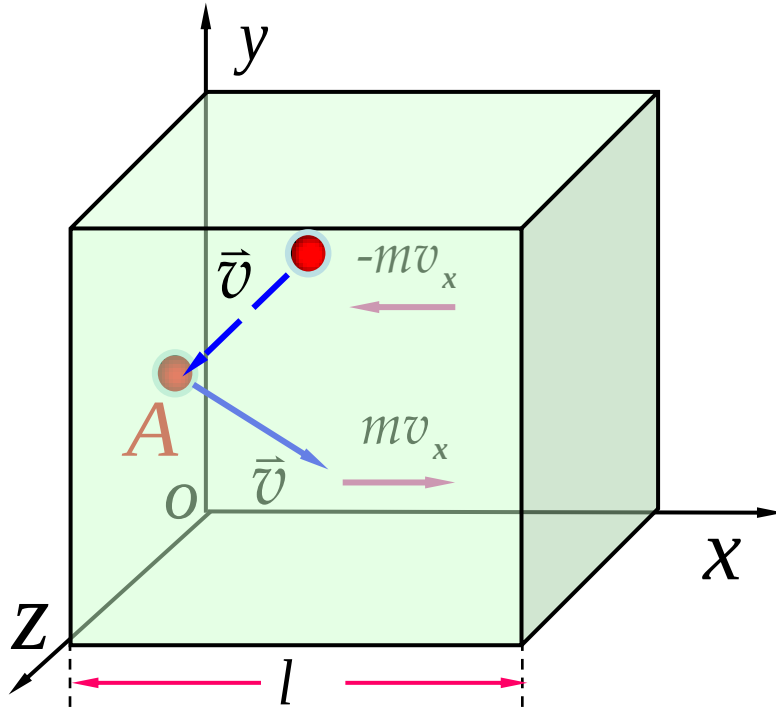
the average value of v^2 for all the molecules

$$\overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2} = \overline{v^2}$$

the motion is completely random, the average values are equal to each other.

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \frac{1}{3} \overline{v^2}$$

2. The molecular interpretation of pressure



Let n moles of an ideal gas be confined in a box of volume V

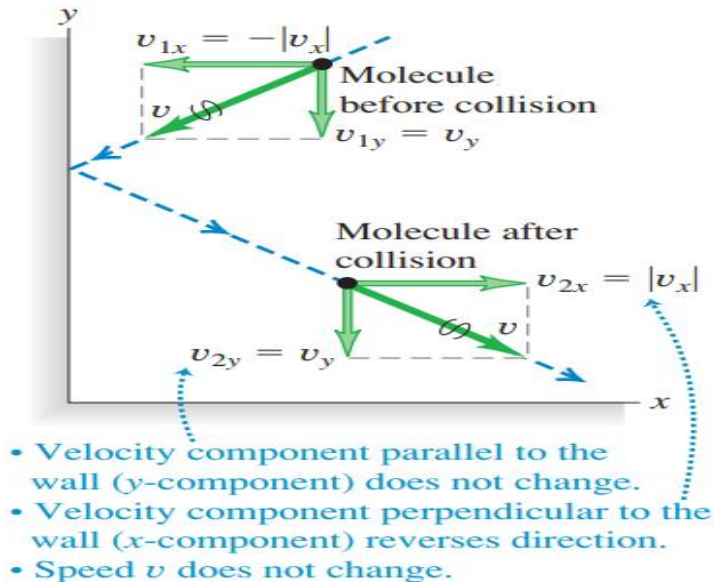
$$p = \frac{F}{A} = \frac{\Delta P}{A \Delta t}$$

Change in momentum of particle i

$$\Delta P_{ix} = 2m v_{ix}$$

Impulse acting on the wall during Δt

$$-2m v_{ix}$$



Time Δt between collisions with the wall

$$\Delta t = 2l / v_{ix}$$

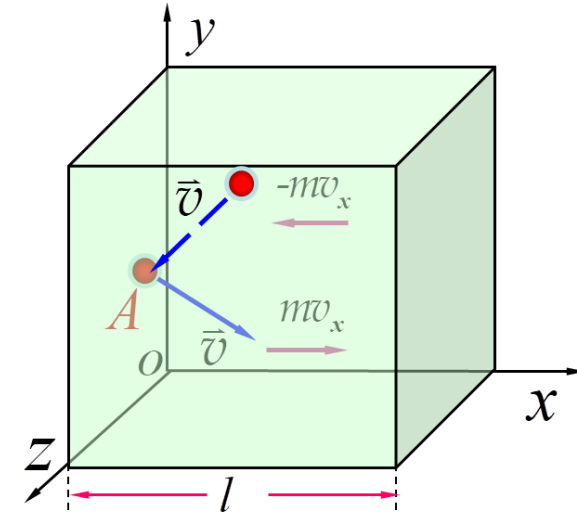
$$F_i = \frac{\Delta P_i}{\Delta t} = \frac{2mv_{ix}}{2l / v_{ix}} = \frac{mv_{ix}^2}{l}$$

The total force exerting the wall

$$F = \sum F_i = \sum \frac{mv_{ix}^2}{l} = \frac{m}{l} \sum v_{ix}^2 = \frac{m}{l} N \overline{v_x^2}$$

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \frac{1}{3} \overline{v^2} \quad F = \frac{m}{l} N \frac{\overline{v^2}}{3}$$

$$p = \frac{F}{A} = \frac{1}{3} \frac{mN \overline{v^2}}{Al} = \frac{1}{3} \frac{N}{V} m \overline{v^2}$$



$$p = \frac{1}{3} \frac{N}{V} m \overline{v^2} = \frac{2}{3} \frac{N}{V} \left(\frac{1}{2} m \overline{v^2} \right) = \frac{2}{3} \frac{N}{V} \bar{K}$$

$$\bar{K} = \frac{1}{2} m \overline{v^2}$$

Average translational kinetic energy of molecules

$$p = \frac{2}{3} \frac{N}{V} \bar{K}$$

$\frac{N}{V}$: 分子密集程度

$\bar{K}$: 分子运动快慢

The pressure is proportional to the number of **molecules per unit volume** and to the **average translational kinetic energy of the molecules**.

Considering collisions with other molecules, will the result change?

Microscopic quantity

$$p = \frac{2}{3} \frac{N}{V} \bar{K}$$

Macroscopic quantity

宏观可测量是微观量的统计平均值

3. Molecular Interpretation of Temperature

compare with the equation of state

$$pV = NkT$$

$$p = \frac{2}{3} \frac{N}{V} \bar{K} \quad \Rightarrow \quad \bar{K} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

Microscopic quantity Macroscopic quantity

The average translational kinetic energy of molecules in an ideal gas is directly proportional to the absolute temperature. Temperature is a direct measure of **average translational molecular kinetic energy**.

温度 T 的物理意义

$$\bar{K} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

(1) 温度是分子平均平动动能的量度.

$$\bar{K} \propto T$$

(2) 温度是大量分子的集体表现.

(3) 在同一温度下各种气体分子平均平动动能均相等.

Question

一瓶氦气和一瓶氮气密度相同，分子平均平动动能相同，而且都处于平衡状态，则：

- (A) 温度相同、压强相同.
- (B) 温度、压强都不同.
- ☒ (C) 温度相同，氦气压强大于氮气压强.
- (D) 温度相同，氦气压强小于氮气压强.

$$P = \frac{N}{V} kT = \rho \frac{k}{m} T$$

Question

理想气体体积为 V ，压强为 p ，温度为 T 。一个分子的质量为 m ， k 为玻耳兹曼常量， R 为摩尔气体常量，则该理想气体的分子数为：

(A) pV/m

(B) $pV/(kT)$

(C) $pV/(RT)$

(D) $pV/(mT)$

解 $p = \frac{N}{V} kT$

$$N = \frac{pV}{kT}$$

Question

(1) 在不改变压强和温度的前提下，把标况下的1mol氮气和2mol氧气混合起来，则氧气分子碰撞器壁产生的压强为

(2) 不改变容器的大小，让氮气和氧气充分发生化学反应合成 NO_2 气体，当反应结束气体温度下降到原始温度时，气体的压强为

(A) 1atm (B) 2atm (c) $1/3$ atm

(D) $2/3$ atm (E) $3/2$ atm

Question

Two **identical rooms** in a house are connected by an **open doorway**. The **temperatures** in the two rooms are maintained at **different values**. Which room contains **more air**?

- A. the room with higher temperature
- ☒ B. the room with lower temperature
- C. the room with higher pressure
- D. neither because both have the same pressure
- E. neither because both have the same volume

Question

关于温度的意义，下列说法正确的是（多选）

- ☒ (A) 气体的温度是分子平均平动动能的量度。
- ☒ (B) 气体的温度是大量气体分子热运动的集体表现，具有统计意义。
- ☒ (C) 温度的高低反应物质内部分子运动剧烈程度的不同。
- ☐ (D) 从微观上看，气体的温度表示每个气体分子的冷热程度。

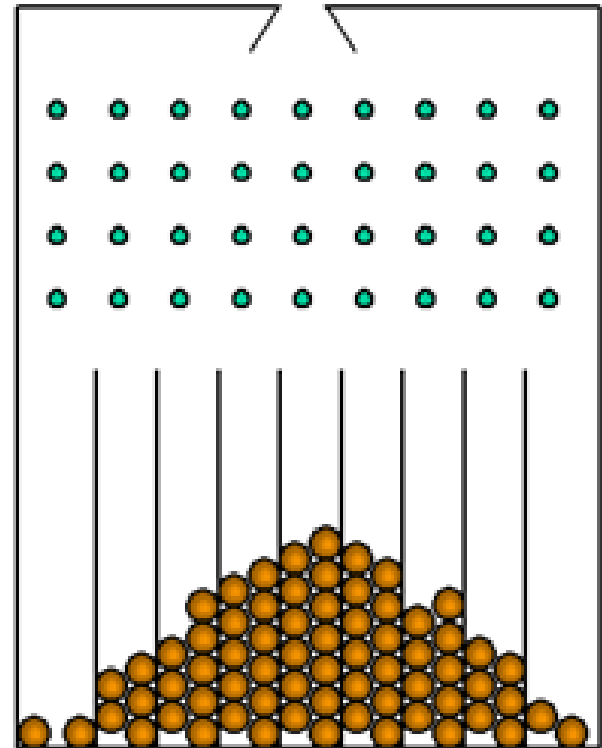
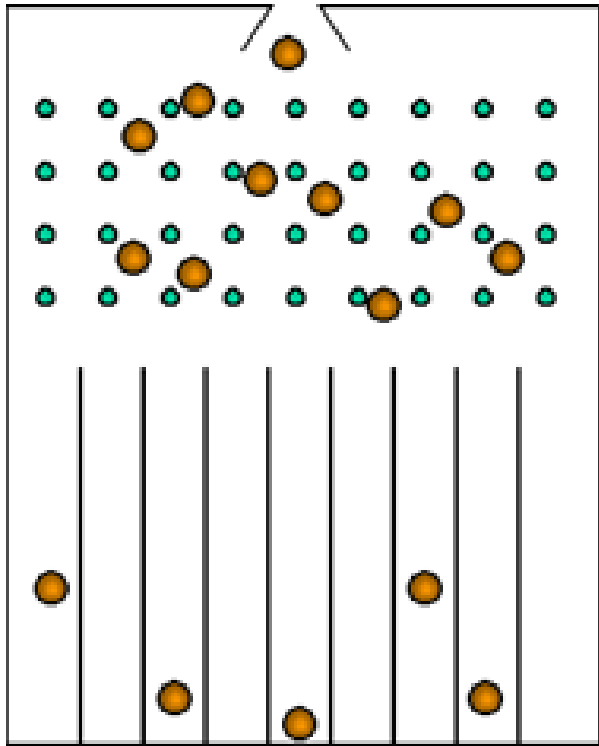
Question

对于一定质量的理想气体，以下说法正确的是

- (A) 如果体积减小，气体分子在单位时间内作用于器壁单位面积的总冲量一定增大
- ☒ (B) 如果压强增大，气体分子在单位时间内作用于器壁单位面积的总冲量一定增大
- (C) 如果温度不变，气体分子在单位时间内作用于器壁单位面积的总冲量一定不变
- (D) 如果压强增大，气体分子在单位时间内作用于器壁单位面积的总冲量一定减小

13.3 Distribution of Molecular speeds

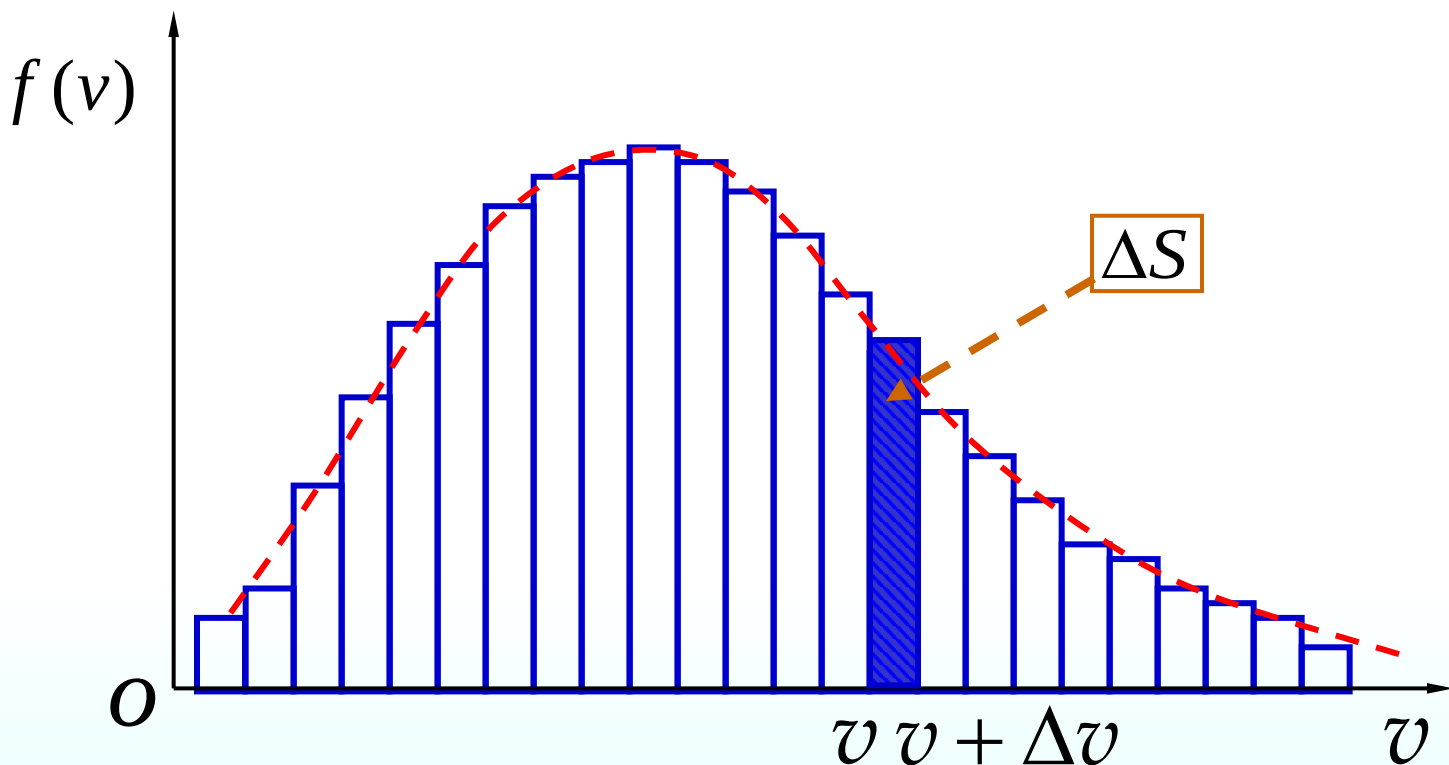
The speed of individual molecule is random, the speeds of immense molecules obey some distribution.



分子速率分布图

N : 分子总数

ΔN : $v \rightarrow v + \Delta v$ 间的分子数.

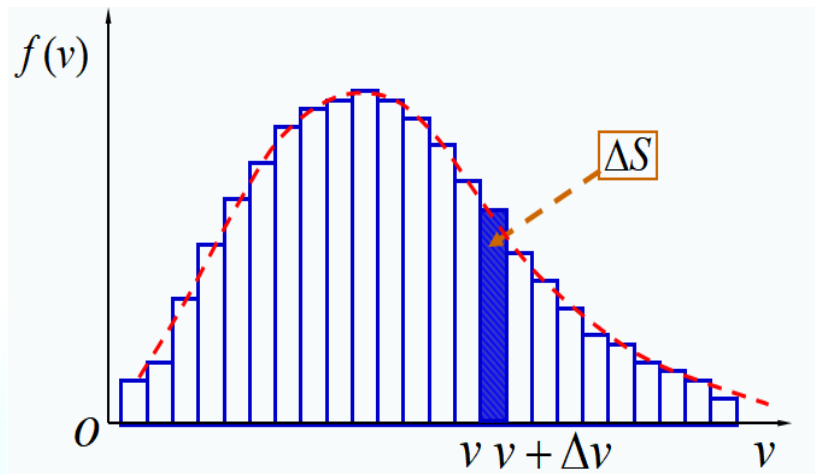


$$\Delta S = \frac{\Delta N}{N}$$

表示速率在 $v \rightarrow v + \Delta v$ 区间的分子数占总数的百分比.

1. The Maxwell distribution (1859):

$$f(v) = 4\pi \left(\frac{M_{mol}}{2\pi RT} \right)^{\frac{3}{2}} v^2 e^{-\frac{M_{mol}v^2}{2RT}}$$

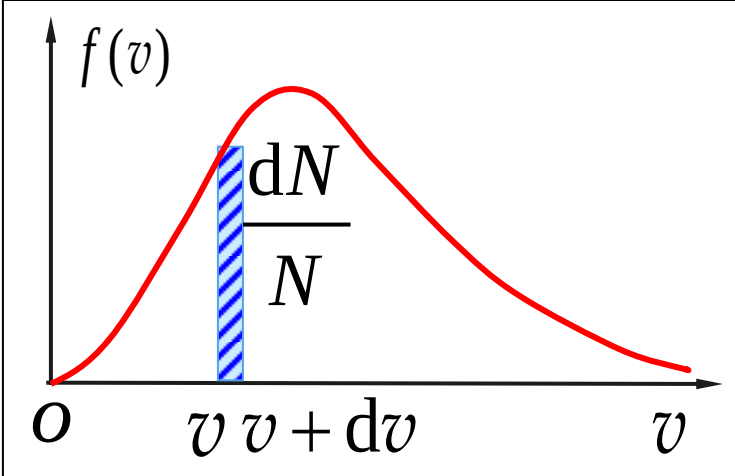


$$f(v) = \frac{\Delta S}{\Delta v} = \frac{\Delta N}{N(\Delta v)}$$

$$\Delta v \rightarrow 0,$$

$$f(v) = \frac{dN}{N(dv)}$$

***f(v)*--probability distribution function: the ratio of the number of molecules that have speed interval 1m/s centered on speed v and total molecules of a gas N .**



$$f(v) = 4\pi \left(\frac{M_{mol}}{2\pi RT} \right)^{\frac{3}{2}} v^2 e^{-\frac{M_{mol}v^2}{2RT}}$$

$$f(v) = \frac{dN}{N(dv)}$$

$$(1) \quad f(v) = \frac{1}{N} \frac{dN}{dv}$$

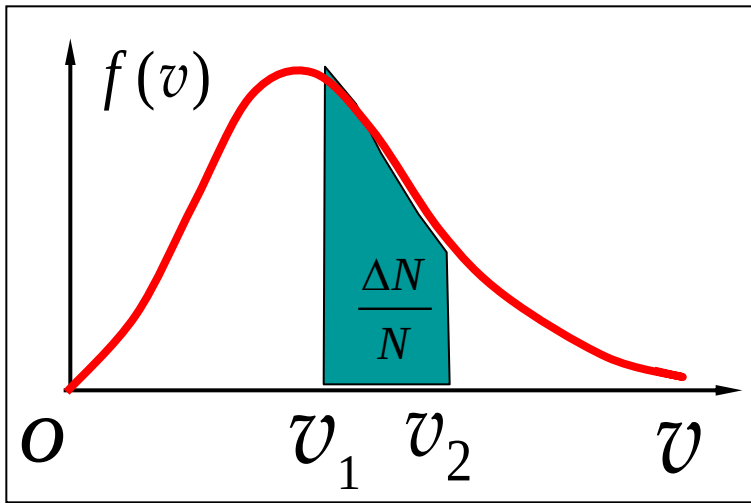
$f(v)$: 速率 v 附近, 单位速率间隔内出现的分子数占总分子数的百分比

$f(v)$: the fraction of molecules with speeds in the interval 1m/s centered on speed v

$$(2) \quad f(v)dv = \frac{dN}{N}$$

$f(v)dv$: 速率 v 附近 dv 速率间隔内出现的分子数占总分子数的百分比

$f(v)dv$: the fraction of molecules with speeds in the interval dv centered on speed v



$$(3) \text{ frac} = \int_{v_1}^{v_2} f(v) dv$$

$v_1 \sim v_2$ 速率区间内分子数占总分子数的百分比

The fraction of molecules with speeds in an interval of v_1 to v_2 .

$$(4) \int_0^{\infty} f(v) dv = 1$$

Normalization 分布函数归一化

Conditions of Maxwell distribution of speeds : thermal equilibrium and bases on the rule of statistics.

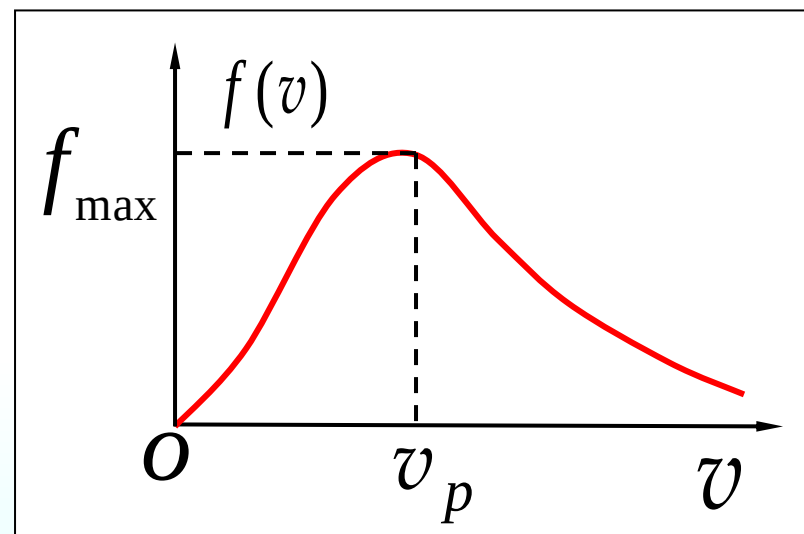
2. Three kinds of special speed

(1) The most probable speed(最概然速率) v_p

(分布在速率 v_p -- v_p+dv 速率间隔的分子数占总分子数的概率最大)

$$\left. \frac{df(v)}{dv} \right|_{v=v_p} = 0$$

$$v_p = \sqrt{\frac{2kT}{m}} \approx 1.41 \sqrt{\frac{kT}{m}}$$



$$\because M_{mol} = mN_A, R = N_A k \quad \therefore v_p \approx 1.41 \sqrt{\frac{RT}{M_{mol}}}$$

$$v_p = \sqrt{\frac{2kT}{m}} \approx 1.41 \sqrt{\frac{RT}{M_{mol}}}$$

The **most probable speed** v_p is the speed at which the distribution curve reaches a **peak**.

气体在一定温度下分布在最概然速率附近单位速率间隔内的相对分子数最多。

(2) The average speed (平均速率) $\bar{v}$

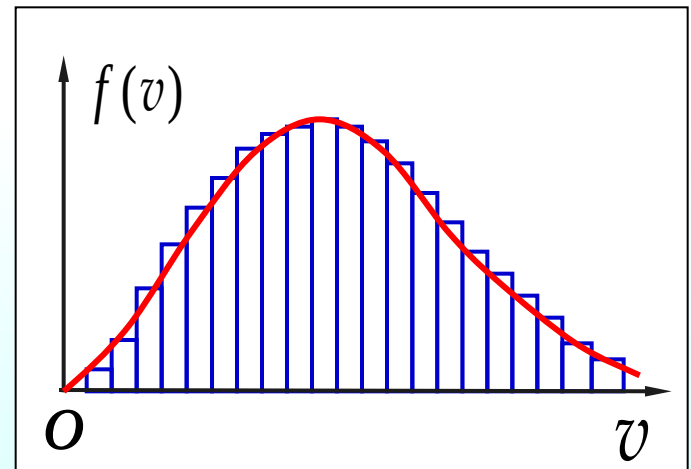
$$\bar{v} = \frac{v_1 dN_1 + v_2 dN_2 + \cdots + v_i dN_i + \cdots + v_n dN_n}{N}$$

$$\bar{v} = \frac{\int_0^N v dN}{N} = \frac{\int_0^\infty v N f(v) dv}{N}$$

$$\int_0^\infty v^3 e^{-av^2} dv = \frac{1}{2a^2}$$

$$\bar{v} = \int_0^\infty v f(v) dv$$

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}} \approx 1.60 \sqrt{\frac{RT}{M_{mol}}}$$



(3) The root-mean-square speed (方均根速率) rms speed

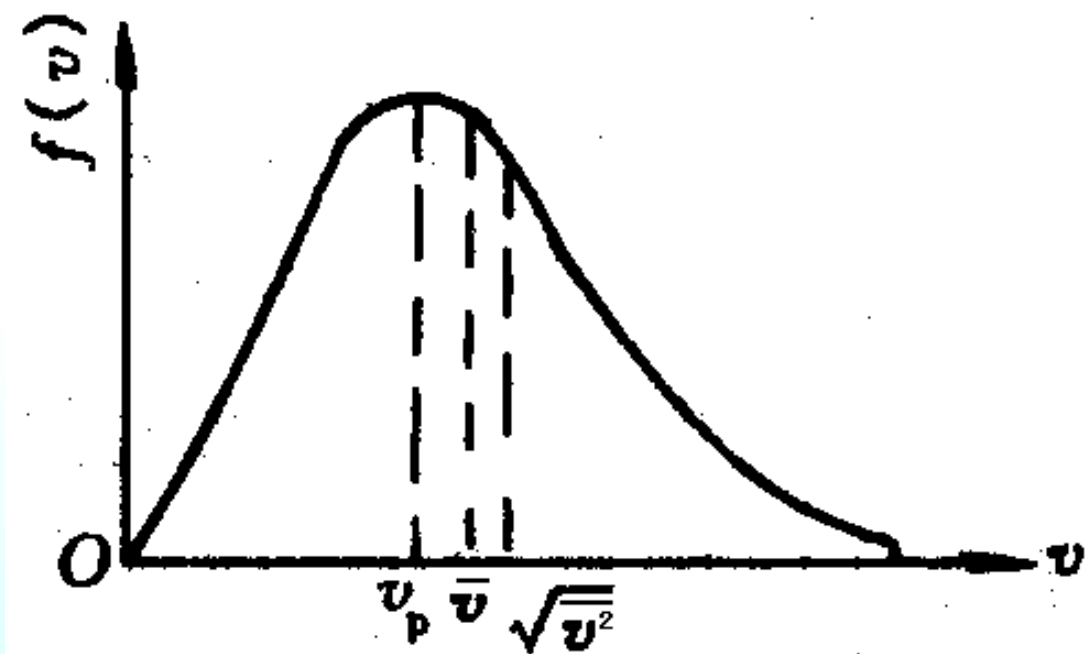
$$\overline{v^2} = \frac{\int_0^N v^2 dN}{N} = \frac{\int_0^\infty v^2 N f(v) dv}{N}$$

$$\overline{v^2} = \int_0^\infty v^2 f(v) dv$$

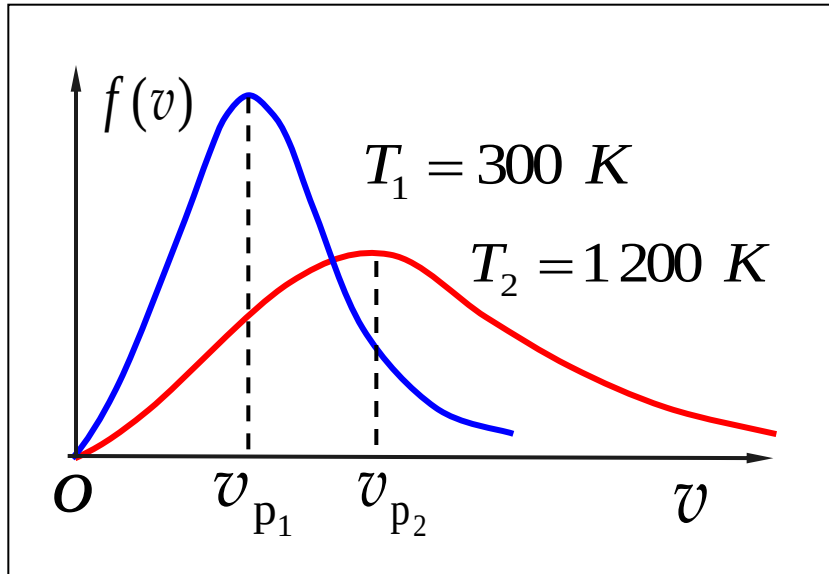
$$\overline{v^2} = 3kT / m$$

$$v_{rms} = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}} \approx 1.73 \sqrt{\frac{RT}{M_{mol}}}$$

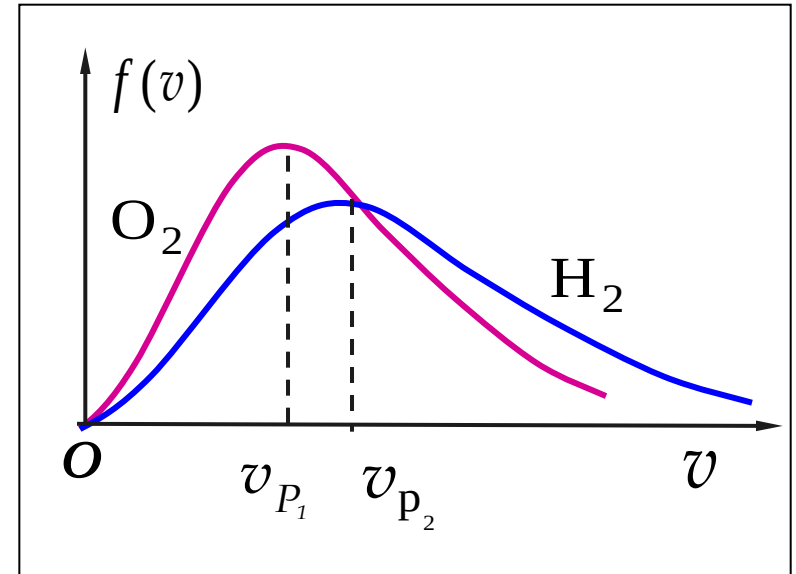
$$v_p < \bar{v} < \sqrt{\overline{v^2}}$$



3 .Two kinds of curves



**the same gas
different temperatures**



**the same temperature
different gases**

Question

真空中的容器内存有等摩尔的氢气和氧气的混合气体，由于容器密闭性不好，有少量气体泄漏。则过一段时间后容器中的氢气和氧气相比（ ）

(A) 更多

☒ (B) 更少

(C) 一样多

Question

A system consists of N gas molecules, each with mass m . Their rms speed is v_{rms} . Their total translational kinetic energy is:

- (A) $\frac{1}{2} m (N v_{rms})^2$
- (B) $\frac{1}{2} N (m v_{rms})^2$
- (C) $\frac{1}{2} m v_{rms}^2$
- ~~(D)~~ $\frac{1}{2} N m v_{rms}^2$
- (E) $N \left(\frac{1}{2} m v_{rms} \right)^2$

Question

According to the Maxwell speed distribution, as the **temperature increases** the number of molecules with speeds within a small interval near the **most probable speed**:

A. increases

☒ B. decreases

C. increases at high temperatures and decreases at low

D. decreases at high temperatures and increases at low

E. stays the same

Question

A sample of argon gas (molar mass 40 g) is at four times the absolute temperature of a sample of hydrogen gas (molar mass 2g). The **ratio(比值) of the rms** speed of the argon molecules to that of the hydrogen is:

(A) 1

(B) 5

(C) $1/5$

(D) $\sqrt{5}$

~~(E)~~ $1/\sqrt{5}$

SP: 已知分子数 N , 分子质量 m , 分布函数 $f(v)$.

求 (1) 速率在 $v_p \sim \bar{v}$ 间的分子数;

(2) 速率在 $v_p \sim \infty$ 间所有分子动能之和 .

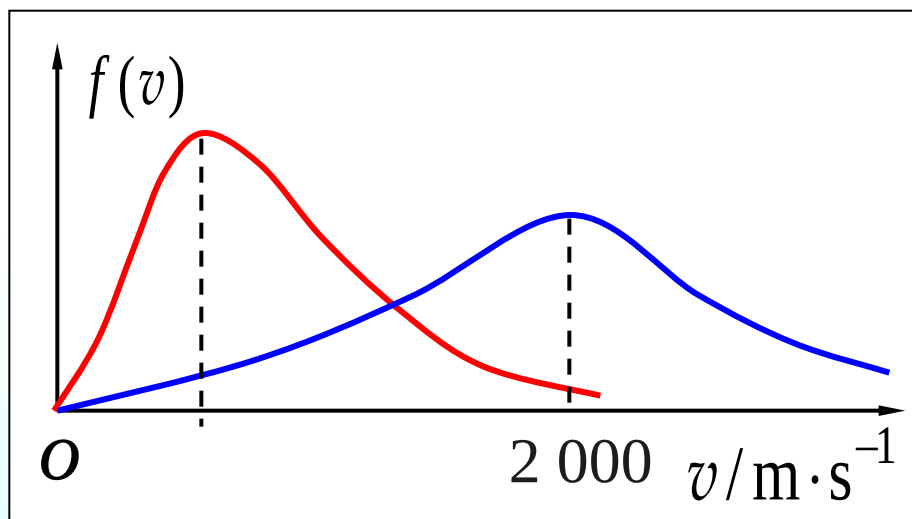
(3) 速率在 $v_p \sim \infty$ 间所有分子平均速率?

解 (1) $\int_{v_p}^{\bar{v}} Nf(v)dv$

(2) $\int_{v_p}^{\infty} (\frac{1}{2}mv^2)Nf(v)dv$

(3) $\frac{\int_{v_p}^{\infty} vNf(v)dv}{N}$ **Wrong!!** $\bar{v} = \frac{\int_{v_p}^{\infty} vNf(v)dv}{\int_{v_p}^{\infty} Nf(v)dv}$

2 如图示两条 $f(v) \sim v$ 曲线分别表示氢气和氧气在同一温度下的麦克斯韦速率分布曲线，从图上数据求出两气体最概然速率。



解 $v_p = \sqrt{\frac{2kT}{m}} \quad \because m(\text{H}_2) < m(\text{O}_2)$

$$\therefore v_p(\text{H}_2) > v_p(\text{O}_2) \quad \therefore v_p(\text{H}_2) = 2\,000 \text{ m.s}^{-1}$$

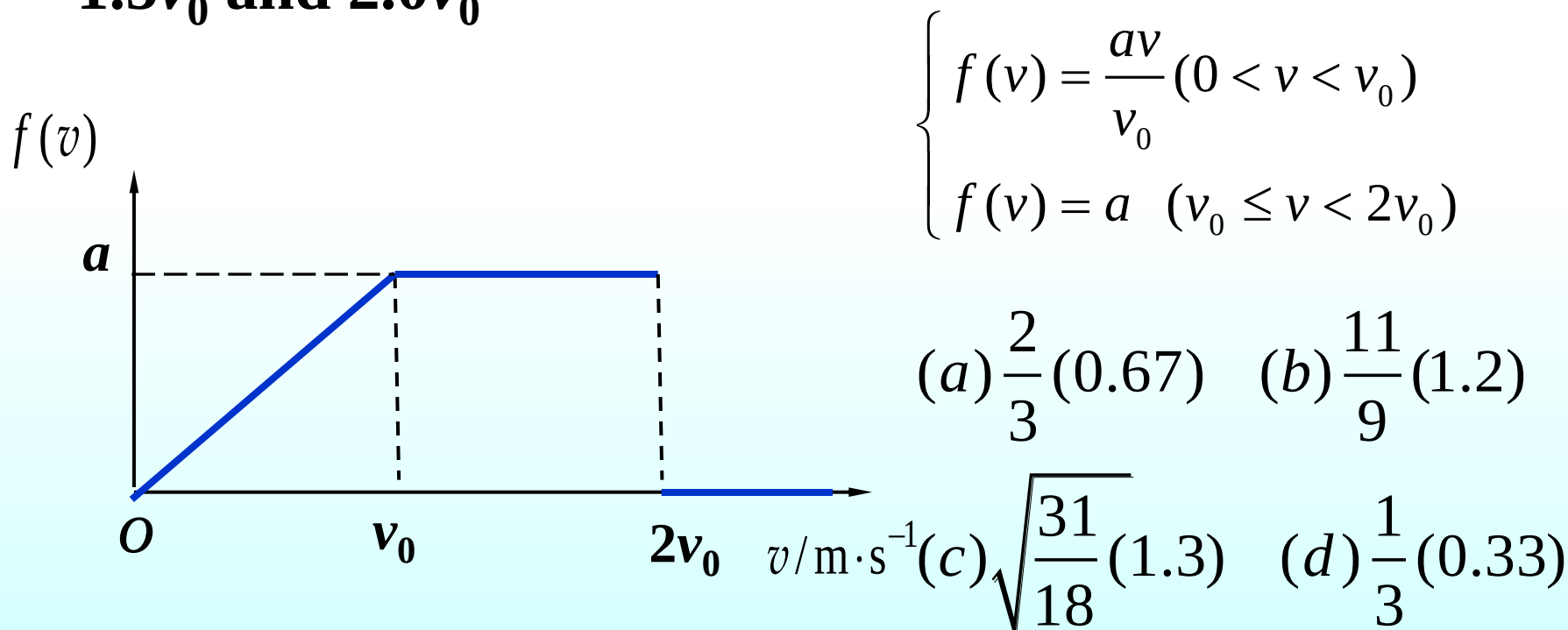
$$\frac{v_p(\text{H}_2)}{v_p(\text{O}_2)} = \sqrt{\frac{m(\text{O}_2)}{m(\text{H}_2)}} = \sqrt{\frac{32}{2}} = 4$$

$$\therefore v_p(\text{O}_2) = 500 \text{ m.s}^{-1}$$

SP :Figure shows a hypothetical speed distribution for a sample of N gas particles (Note that $f(v)=0$ for speed $v > 2v_0$) what are the values of

(a) av_0 **(b)** v_{avg} / v_0 **(c)** v_{rms} / v_0

(d) What fraction of the particles has a speed between $1.5v_0$ and $2.0v_0$



Question

A quantity of gas containing N molecules has a speed distribution function $f(v)$. **How many** molecules have speeds between v_1 and v_2 ($v_2 > v_1$)?

(A) $\int_0^{v_2} f(v)dv - \int_0^{v_1} f(v)dv$

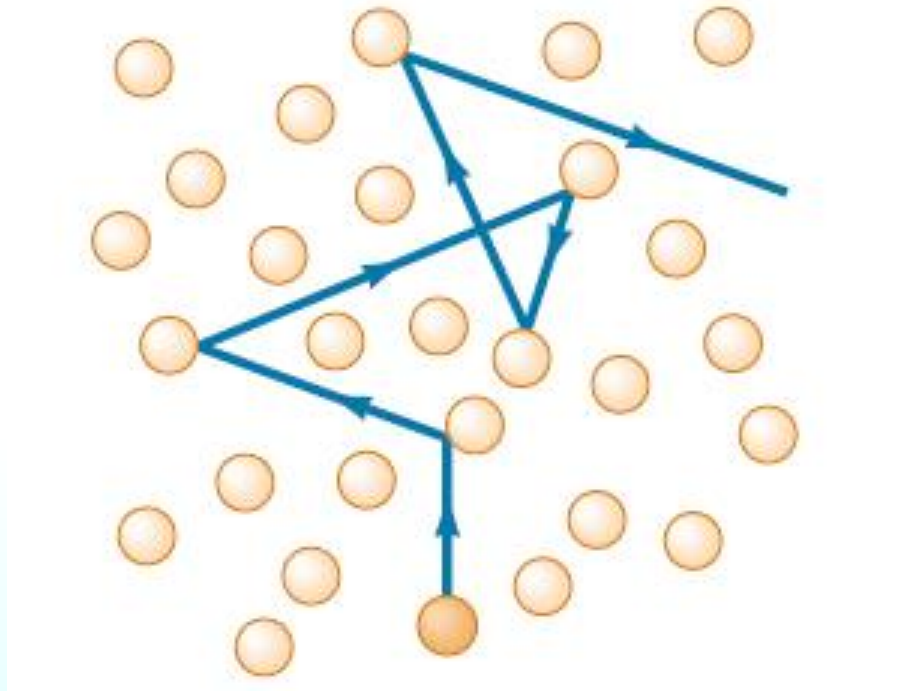
(B) $N[\int_0^{v_2} f(v)dv - \int_0^{v_1} f(v)dv]$

(C) $\int_0^{v_1} f(v)dv - \int_0^{v_2} f(v)dv$

(D) $N[\int_0^{v_1} f(v)dv - \int_0^{v_2} f(v)dv]$

* 13.4 Mean free path (平均自由程)

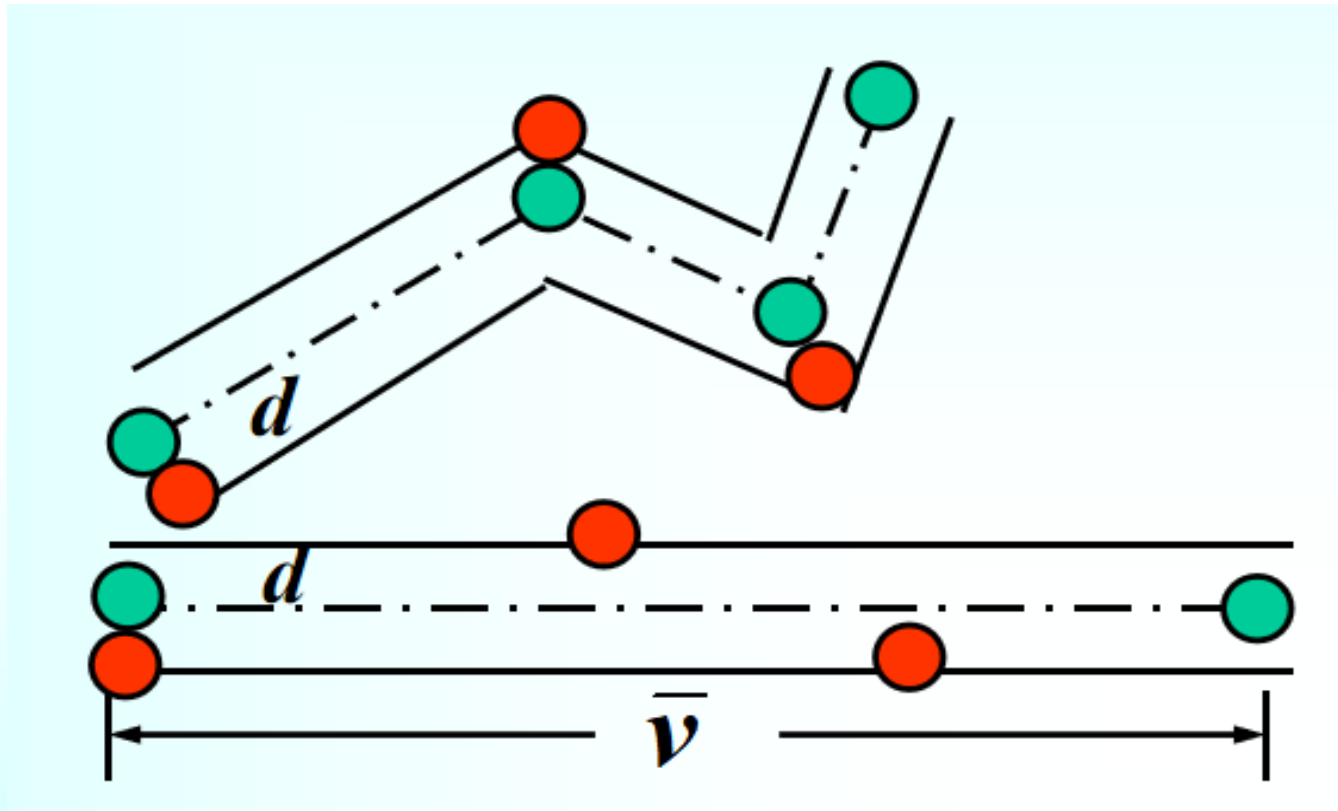
If we were to follow the path of a particular molecule, we would expect to see it follow a **zigzag** path



A molecule moving through a gas collides with other molecules in a random fashion. This behavior is sometimes referred to as *a random-walk process*.

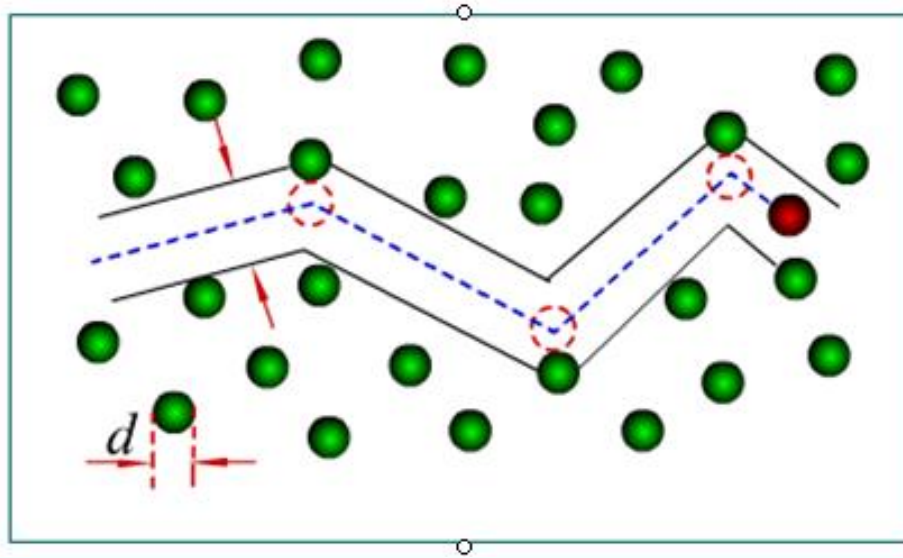
Mean free path: the average distance a molecule travels between two **adjacent** collisions

每两次连续碰撞之间，一个分子自由运动的平均路程。



Suppose:

- (1) Molecules are hard spheres of radius r (有效半径)**
- (2) other molecules are still**



In a time Δt

The **number of collisions (碰撞次数)**

$$\pi (2r)^2 \bar{v} \Delta t \left(\frac{N}{V} \right) = \pi (2r)^2 \bar{v} \Delta t \left(\frac{N}{V} \right)$$

Distance traveling $\bar{v} \Delta t$

Mean free path

$$l_M = \frac{\bar{v} \Delta t}{(\frac{N}{V}) \pi (2r)^2 \bar{v} \Delta t} = \frac{1}{(\frac{N}{V}) \pi (2r)^2} = \frac{1}{4\pi r^2 (N/V)}$$

Other molecules are moving

$$v_{rel} = \sqrt{2}\bar{v}$$

$$l_M = \frac{1}{4\pi r^2 \sqrt{2} (N/V)}$$

Collision frequency f : 单位时间内一个分子和其它分子碰撞的平均次数.

$$f = \frac{\bar{v}}{l_M} = 4\pi \sqrt{2} r^2 (N/V) \bar{v}$$

$$l_M = \frac{1}{4\pi\sqrt{2}r^2(N/V)}$$

$$p = \frac{N}{V} kT$$

$$l_M = \frac{kT}{4\pi\sqrt{2}r^2 p}$$

$$T \text{ 一定时} \quad l_M \propto \frac{1}{p}$$

$$p \text{ 一定时} \quad l_M \propto T$$

If you try to walk through a crowd, your **mean free path**—the distance you can travel on average without running into another person—depends on how large the people are and how closely they are spaced.



Mean free path of air molecules at STP.

Estimate the mean free path of air molecules at STP, standard temperature and pressure (0°C, 1atm). The diameter of O₂ and N₂ molecules is about 3×10⁻¹⁰ m.

Solution:

$$\frac{N}{V} = \frac{6.02 \times 10^{23}}{22.4 \times 10^{-3}} = 2.69 \times 10^{25} \text{ molecules/m}^3$$

$$l_M = \frac{1}{4\pi\sqrt{2}r^2(N/V)}$$

$$l_M = \frac{1}{4\pi\sqrt{2}(1.5 \times 10^{-10})^2(2.7 \times 10^{25})} \approx 9 \times 10^{-8} \text{ m}$$

SP: 一容器内储有氧气，温度为 27°C ，其压强为 $1.02 \times 10^5 \text{Pa}$ ，求：(1) 气体分子数密度；(2) 氧气的密度；(3) 分子的平均平动动能；(4) 分子间的平均距离。

解 (1) $\frac{N}{V} = p / kT = 2.44 \times 10^{25} \text{m}^{-3}$

$$(2) \rho = M / V = pM_{\text{mol}} / RT = 1.3 \text{ kg} \cdot \text{m}^{-3}$$

$$(3) \bar{K} = 3kT / 2 = 6.21 \times 10^{-21} \text{ J}$$

$$(4) \bar{d} = \sqrt[3]{1/(N/V)} = 3.45 \times 10^{-9} \text{ m}$$

Question

设有一**恒温**容器，其内储有某种理想气体，若容器发生缓慢漏气，问

(1) 气体的压强是否变化？为什么？

(2) 容器内气体分子的平均平动动能是否变化？为什么？

解

$$pV = nRT \quad n \downarrow \Rightarrow p \downarrow \quad \overline{K} = \frac{3}{2}kT \text{ 不变}$$

Question

理想气体在等压过程中的平均碰撞频率与温度 T 的关系

(A) 与 T 无关

(B) $f \propto \sqrt{T}$

(C) $f \propto 1/\sqrt{T}$

(D) $f \propto T$

$$l_M = \frac{1}{4\pi\sqrt{2}r^2(N/V)}$$

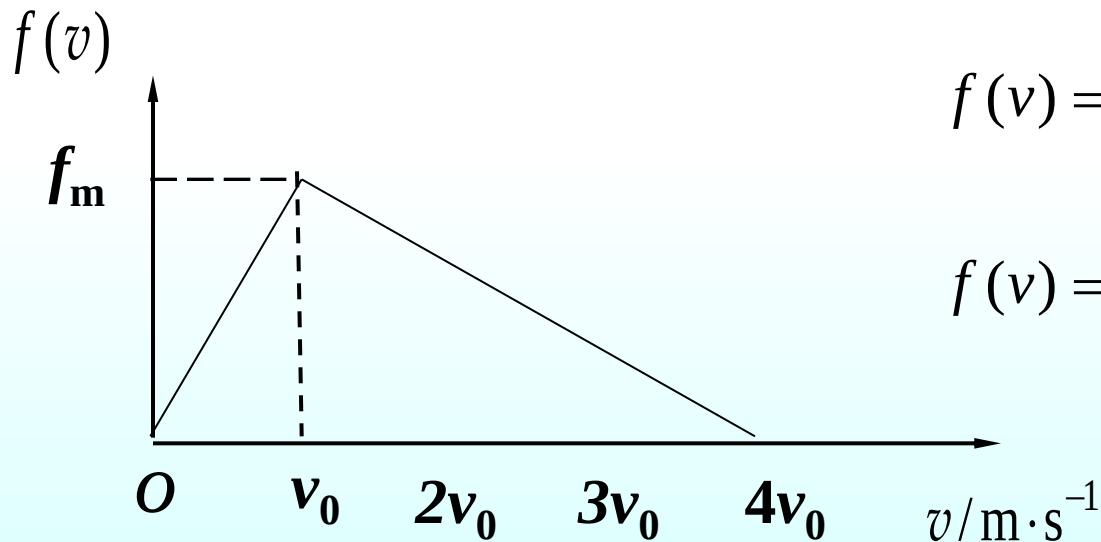
$$f = \frac{\bar{v}}{l_M} = 4\pi\sqrt{2}r^2(N/V)\bar{v} = 4\pi\sqrt{2}r^2 \frac{P}{kT} \sqrt{\frac{8kT}{\pi m}} \propto \frac{1}{\sqrt{T}}$$

SP: 设想 N 个分子组成的理想气体分子的速率分布曲线简化成如图所示的形状, 其中 v_0 已知, 求

(1) 速率分布函数的极大值 f_m $1/2v_0$

(2) $0.5v_0 \sim 2v_0$ 速率区间内的分子数; $\Delta N = \frac{29}{48} N$

(3) N 个分子的平均速率。 $\bar{v} = \frac{5}{3} v_0$



$$f(v) = \frac{v}{2v_0^2} (0 < v < v_0)$$

$$f(v) = -\frac{v}{6v_0^2} + \frac{2}{3v_0} (v_0 < v < 4v_0)$$